

SUGEERTH MURUGESAN

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Education

University of California, Davis

Davis, CA

Ph.D. in Computer Science – Visualization & Machine Learning; GPA: 3.91/4.0

2013 – 2017

- Research Affiliate at UC Berkeley; Thesis: Interactive Methods for Deep Learning & Graph Data
- Coursework: Neural Networks, Statistical Learning, NLP, Probabilistic Graphical Models, Distributed Systems

Amrita University & UC Davis

India & Davis, CA

B.Tech. in Computer Science – Top 1% Exchange Scholar

2009 – 2013

Professional Experience

Intuit

Oct 2024 – Present

Staff AI Scientist – ML/AI R&D

Mountain View, CA

- Developed **5+ production ML models end-to-end**—from problem formulation and data collection through training, evaluation, and deployment—powering personalization for **5M+ TurboTax users**
- Drove **10x improvement in revenue metrics** by designing recommendation models with novel feature engineering; owned full **ML lifecycle** from experimentation to A/B testing to production rollout
- Architected **next-gen feature store infrastructure** using Tecton, improving feature freshness by **3x** and reducing feature engineering time by **40%** across 10+ ML models
- Built **ML engineering pipelines** (Spark, Kafka, Airflow) for model training and serving; established **CI/CD for ML** with automated retraining, model validation, and canary deployments
- Led research on **LLM-powered recommendation systems**; prototyped and deployed agentic AI workflows using LangGraph for complex multi-step reasoning tasks

Walgreens AI – Office of CTO

Aug 2023 – June 2024

Senior Data Scientist – ML Infrastructure Lead

Bay Area, CA & Chicago, IL

- Led team of 3 data scientists building **ML data pipelines** for document processing at scale; designed **feature extraction systems** processing **16M+ records** with **98% accuracy**
- Built **model evaluation infrastructure** with automated benchmarking, A/B testing frameworks, and **performance monitoring dashboards** using Grafana and custom metrics
- Designed **data curation workflows** for training data quality; implemented **human-in-the-loop labeling pipelines** that improved downstream model accuracy by **35%**
- Optimized **Spark-based feature pipelines** reducing compute costs by **24%** while maintaining **low-latency serving** for production inference

Intel Corporation – Advanced Applications

Feb 2020 – Aug 2023

Staff Machine Learning Engineer – Platform Tech Lead

Santa Clara, CA

- Led **ML platform team of 4 engineers**; drove 10+ R&D cycles building **DeepInsight**, a distributed ML infrastructure for real-time feature computation and model serving
- Architected **streaming feature pipelines** using **Kafka and Flink** for near-real-time ML applications; achieved **sub-100ms latency** for high-throughput inference serving millions of requests/day
- Built **centralized feature store** enabling feature sharing across 5+ ML domains; implemented **feature discovery** and versioning that accelerated experimentation by **45%**
- Designed **model evaluation infrastructure** with automated metrics collection, **anomaly detection**, and alerting; improved model reliability through comprehensive observability
- Led platform expansion across multi-node **distributed systems on AWS**; established engineering best practices in **Python, Scala, and Go** for team of 3+ staff members

Samba TV – Office of CTO (Time Warner, Disney Funded)

Aug 2018 – Jan 2020

Research Engineer – ML Infrastructure

San Francisco, CA

- Built **high-throughput ML inference system** serving ensemble of 3+ models (logo detection, OCR, prediction) processing **millions of video frames/day** with **F1-score of 82.2%**
- Optimized **feature serving infrastructure** with in-memory store holding **4B+ fingerprints** using RocksDB; improved data loading performance by **32.4%** and reduced lock contention by **14%**
- Enhanced **real-time ACR pipeline** using LSTM models in **C++ and Python**; achieved **21.2% speedup** in detection latency for production serving

Adobe Research

Jan 2018 – Apr 2018

Data Science Research Intern

San Jose, CA

- Built **model evaluation tools** for comparing deep learning architectures; developed visual analytics for **model performance comparison** across 40K+ test cases
- Published **DeepCompare** (IEEE CG&A): frameworks for **systematic model evaluation** and performance analysis integrated into Adobe Sensei platform

Lawrence Berkeley National Laboratory

Mar 2015 – Dec 2017

Graduate Student Researcher

Berkeley, CA

- Led research team developing **evaluation frameworks for ML systems**; published 5+ papers with **150+ citations**; won **BrainKDD Best Paper Award**
- Created **open-source Python libraries** for data processing and model evaluation used by **500+ researchers**; built benchmark datasets for validating model accuracy
- Developed **GPU-accelerated data pipelines** for large-scale scientific computing using **CUDA**; optimized batch processing achieving **22% improvement** in throughput

Technical Skills

ML Infrastructure: Feature Stores (Tecton), ML Pipelines, Model Serving, Feature Engineering, Data Curation, Model Evaluation

Data Processing: Apache Spark, Apache Flink, Apache Kafka, Databricks, ETL Pipelines, Streaming Systems

Languages: Python, Scala, SQL, C++, Go, Rust, R, JavaScript, CUDA

ML/AI Frameworks: PyTorch, TensorFlow, JAX, HuggingFace, LangChain, MLFlow, KubeFlow

Cloud & Infrastructure: AWS (SageMaker, EC2, S3, EMR), Kubernetes, Docker, TensorRT, Triton Inference Server

Databases & Storage: RocksDB, Neo4j, MongoDB, PostgreSQL, Vector Stores, GraphQL

Observability & Tools: Grafana, Prometheus, Jupyter/Notebooks, FastAPI, Git

Selected Publications

DeepCompare: Visual Methods for Model Evaluation and Performance Analysis

IEEE CG&A 2018

Netostat: Analyzing Dynamic Flow Patterns in High-Speed Networks

Cluster Computing 2022

Brain Modulyzer: Interactive Visual Analysis of Functional Brain Connectivity

IEEE TCBB 2016

BrainKDD Best Paper: Hierarchical Spatio-Temporal Visual Analysis of Cluster Evolution

ACM-BCB 2016

Awards & Leadership

2016 – Best Paper Award, BrainKDD at ACM-BCB Conference

2015-2017 – Graduate Fellow, Lawrence Berkeley National Laboratory

2014-2017 – UC Davis Scholar Graduate Fellowship

Technical Leadership – Led teams of 3-4 engineers across multiple organizations; mentored 5+ junior engineers

Reviewer – ICML, NeurIPS, ACL, IEEE VIS, ACM CHI, RecSys, IEEE TVCG